

2. A. Using a suitable regression dataset, implement Linear Regression using scikit-learn.

CODE A:

```
#Practical_7b
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
print("Libraries Loaded.")
#Load dataset
data = pd.read_csv("student_marks.csv")
print("Dataset Loaded.")
#Input and output
X = data[['Hours']]
Y = data[['Marks']]

#Split dataset
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.3, random_state=1)
print("Dataset splitted successfully.")
#Train model
model = LinearRegression()
model.fit(X_train, Y_train)
print("Model Trained Successfully.")
#Display accuracy
print("Training Accuracy:",
      round(model.score(X_train, Y_train) * 100, 2))
print("Testing Accuracy: ",
      round(model.score(X_test, Y_test) * 100, 2))
new_data = pd.DataFrame({'Hours': [11]})

pred = model.predict(new_data)

print("Predicted Marks for 11 Hours:", round(pred.item(), 2))
```

OUTPUT:

Libraries Loaded.

Dataset Loaded.

Training Accuracy: 99.64

Dataset splitted successfully.

Testing Accuracy: 98.71

Predicted Marks for 11 Hours: 93.57

2. B. California Housing Dataset using Linear Regression

Problem Statement : A real estate company wants to predict the Median House Value based on housing features.

Target: Median House Value.

Tasks

1. Load the dataset using Pandas.
2. Display the first five records.
3. Check for missing values.
4. Select Median Income as the independent variable.
5. Split the dataset into training and testing sets.
6. Train a Linear Regression model.
7. Predict the house prices.
8. Display:
 - Coefficient
 - Intercept
 - R^2 Score
9. Plot the regression line.
10. Interpret the results.

CODE:

```
#Practical 2(code b)
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
print("Libraries Loaded Successfully.")

# Step 1: Load Dataset
data = pd.read_csv("housing.csv")
print("Dataset Loaded Successfully.\n")

# Step 2: Display First Five Records
print("First Five Records")
print(data.head())

# Step 3: Check Missing Values
print("\nMissing Values")
print(data.isnull().sum())

# Remove Missing Values
data = data.dropna()

# Step 4: Select Independent and Dependent Variables
X = data[['median_income']]
Y = data['median_house_value']

# Step 5: Split Dataset
X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y,
    test_size=0.3,
    random_state=42
)

print("\nDataset Split Successfully.")

# Step 6: Train Linear Regression Model
model = LinearRegression()
model.fit(X_train, Y_train)

# Step 7: Predict House Prices
```

```

Y_pred = model.predict(X_test)

# Step 8: Display Results
print("\nCoefficient :", model.coef_[0])
print("Intercept :", model.intercept_)
print("R2 Score :", r2_score(Y_test, Y_pred))

# Step 9: Plot Regression Line
plt.figure(figsize=(8,6))

plt.scatter(X_test, Y_test, color='blue', label='Actual Data')

plt.plot(X_test, Y_pred, color='red', linewidth=2, label='Regression Line')

plt.title("Median Income vs Median House Value")
plt.xlabel("Median Income")
plt.ylabel("Median House Value")
plt.legend()

plt.show()

```

OUTPUT:**Libraries Loaded Successfully.****Dataset Loaded Successfully.****First Five Records**

	longitude	latitude	...	median_house_value	ocean_proximity
0	-122.23	37.88	...	452600	NEAR BAY
1	-122.22	37.86	...	358500	NEAR BAY
2	-122.24	37.85	...	352100	NEAR BAY
3	-122.25	37.85	...	341300	NEAR BAY
4	-122.25	37.85	...	342200	NEAR BAY

Missing Values

longitude	0
latitude	0
housing_median_age	0
total_rooms	0
total_bedrooms	207
population	0
households	0
median_income	0
median_house_value	0
ocean_proximity	0
dtype:	int64

Coefficient : 41690.33611910387**Intercept : 45596.46754083503****Dataset Split Successfully.****R2 Score : 0.4761911795039646**

